BASERA

A

SYNOPSIS REPORT OF PROJECT

Basera For the course **FRONT END WEB DEVELOPMENT (26CA205PCB)**

In partial fulfillment of requirements for the degree of

MASTER OF COMPUTER APPLICATIONS (MCA)

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Abstract

Basera is a student-focused web application designed to simplify the discovery and comparison of PGs, flats, hostels and rooms near colleges and universities. It brings scattered accommodation information into one structured platform, allowing students to search by college or locality, filter listings, view property details, save and compare options, and submit enquiries based on rent, deposit, distance, room type, food, Wi-Fi, AC, ratings and other amenities. The project will be developed in two phases: in Semester 1 HTML5, CSS3, JavaScript and React.js frontend using sample data and browser-side storage, followed by a Semester 2 MERN-stack application (MongoDB, Express.js, Node.js) with persistent data, authentication, owner/admin functions and REST APIs. Basera aims to reduce repetitive searching and make student accommodation decisions simpler and more informed.

Motivation

- **Problem Identification:** Students relocating for college often lack information about nearby localities, rent ranges, room types and facilities, and must manually compare multiple sources.

- **Student-First Approach:** Unlike general real-estate platforms, Basera focuses on student-relevant factors in a consistent, comparable format.

- **Academic & Real-World Value:** Addresses a recurring real-world problem while remaining practical as an academic project.

- **Value to Users & Owners:** Reduces fragmented searching for students and gives owners a structured channel to reach them.
- **Alignment with Emerging Tech:** Final system uses the MERN stack and can be extended with maps and recommendation features.

Introduction

Basera belongs to the property-tech and student-services domain and is designed as a web platform for discovering and comparing PGs, flats, hostels and rooms. Each listing provides shortlisting information such as location, rent, deposit, type, distance, amenities, ratings, description and enquiry options. The Semester 1 version is a frontend website using React.js ; backend and database functionality will follow in Semester 2 using Node.js/Express.js and MongoDB, supported by VS Code, Git/GitHub, Figma, Postman and a maps/location service.

Objectives

1. Design a responsive, easy-to-understand interface for accommodation discovery.
2. Allow search by college, locality, area or property information.
3. Provide filters for budget, accommodation type, room type, rating and amenities.
4. Present property information in a consistent, readable format.
5. Enable saving properties to a favourites/shortlist section.
6. Enable side-by-side comparison of selected properties.
7. Provide a student–property enquiry flow (full-stack phase).
8. Provide a structured property-listing flow for owners.
9. Create a frontend foundation extendable into a full MERN application in Semester.

Future Scope

- AI-based recommendations using budget, location, room type and amenities.
- Roommate matching based on compatible preferences and lifestyle.
- Map-based search, travel-distance information and nearby-college discovery.
- Property/owner verification workflows and reporting of suspicious listings.
- Area-wise rental trends and estimated cost insights; multilingual and mobile app support.
- Notifications for new/updated listings; booking or payment integration after verification and legal review.

Findings

The main challenge is not just displaying listings but helping students evaluate them consistently across rent, deposit, distance, food, Wi-Fi, AC, sharing, security and reviews — making a comparison-oriented interface central to the design. Claims such as “verified” or “available now” require reliable data sources; the Semester 1 prototype therefore uses sample data, while the final system will need database-backed records, owner/admin workflows and validation.

Analysis Phase

The analysis phase defines the users, requirements, information flow and boundaries of Basera. The three planned roles are Student/User, Property Owner and Administrator. The frontend prototype mainly demonstrates the student journey and owner-listing interface, role-based authentication and admin controls are planned for the backend phase.

Requirement Analysis – Functional

• Search accommodation by college, locality, area or property name.

• Display PG, flat, hostel and room listings with filtering (rent, type, room type, rating, amenities).

• Provide a detailed view of each property; add/remove favourites; compare properties side by side.

• Provide an enquiry action for a selected property.

• Owner interface to submit property details (persistent create/edit in backend phase).

• Full-stack version: authentication, role-based access, and persistent data via MongoDB.

Requirement Analysis – Non - Functional

• **Usability:** understandable navigation and information for first-time users.

• **Responsiveness:** adapts to desktop, tablet and mobile screens.

• **Performance & Reliability:** efficient search/filter response; graceful error handling.

• **Security:** secure authentication, authorization, validation and password handling (backend phase).

Data Flow Diagrams:

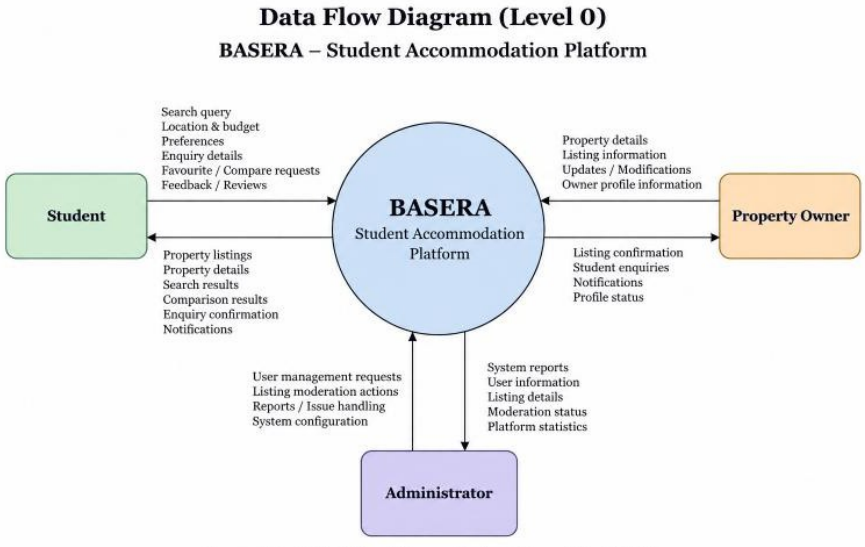


Figure 1: Level 0 Data Flow Diagram (Context Diagram) for BASERA

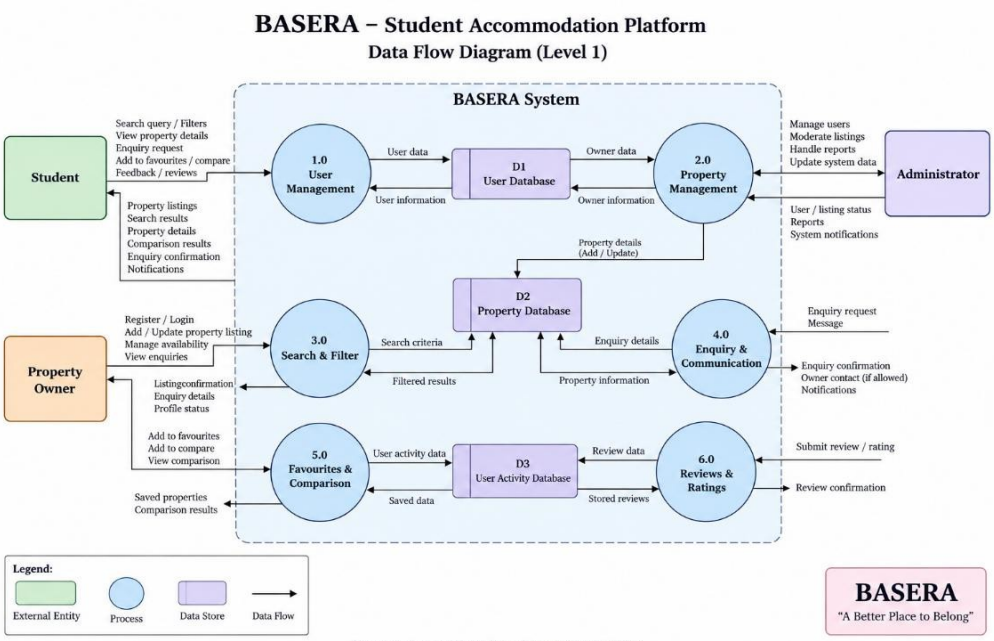


Figure 2: Level 1 Data Flow Diagram for BASERA

Technology Stack

Area	Technology	Purpose
Frontend (Sem 1)	HTML5, CSS3, JavaScript	Structure, styling, responsive prototype
Frontend (Final)	React.js (MERN)	Components, routing, dynamic state
Backend (Sem 2)	Node.js + Express.js	REST APIs, auth, validation, logic
Database (Sem 2)	MongoDB	Persistent users, properties, reviews, enquiries
Design	Figma	Wireframes and UI/UX planning
Development	VS Code	Source-code development
Version Control	Git + GitHub	Version control and collaboration
API Testing	Postman	Testing backend endpoints (Sem 2)
Deployment	Vercel/Netlify + Node hosting	To be finalized during implementation
Location Service	Maps/Location API (planned)	Map display and distance functionality

System Design & Methodology

Basera follows an incremental Agile-style development approach, implemented and tested module by module. The design separates the user interface, application logic and data layer.

Semester 1 uses sample data and browser-side state to demonstrate the complete frontend journey; Semester 2 replaces these with REST APIs and MongoDB persistence.

1. **Requirement Analysis**: Identify user roles, problems, functional requirements and project boundaries.

2. **UI/UX Design**: Navigation, page structure, listing cards, filters, property details, comparison and forms.

3. **Frontend Development**: Responsive pages/interactions in HTML, CSS, JS and React.js for MERN frontend.

4. **Backend APIs**: Node.js/Express.js endpoints for authentication, properties, favourites, reviews, enquiries.

5. **Database Integration**: Design MongoDB collections and connect APIs to persistent data.

6. **Testing**: Functional, form-validation, responsive, API, database, authentication and error-handling tests.

7. **Deployment:** Deploy frontend and backend on suitable hosting; verify end-to-end application.

Expected Outcomes

Semester 1 delivers a responsive frontend prototype demonstrating the complete Basera user journey; Semester 2 delivers a functional MERN web application with persistent data and server-side functionality:

• A responsive, multi-device compatible student accommodation website/web application.

• Searchable and filterable accommodation listings with detailed property pages.

• Working favourites and property-comparison flows.

• An owner property-submission interface and student enquiry flow.

• Integration of React frontend, Node/Express backend and MongoDB database in the final phase.

• Role-based functionality for students, owners and administrators; deployment on a live/approved server.

References

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4. Express.js Documentation – https://expressjs.com/
5. MongoDB Documentation – https://www.mongodb.com/docs/
6. Git and GitHub Documentation – https://git-scm.com/docs
7. Figma Documentation – https://developers.figma.com/docs/plugins/api/api-reference/
8. Course material and faculty-provided Front End Web Development synopsis format